

ISYE 6420 Homework 4

Question 1

1a)

Setting up probability density distributions.

Likelihood:

$$L(X, \theta, \mu) = \sqrt{\frac{\theta}{2\pi}} e^{-\frac{\theta(x)^2}{2}}$$

Likelihood at $x = -2$:

$$L(X, \theta, \mu) = \sqrt{\frac{\theta}{2\pi}} e^{-2\theta}$$

Prior and proposal distributions are Gamma distributions.

Prior $\sim Ga(1/2, 1)$:

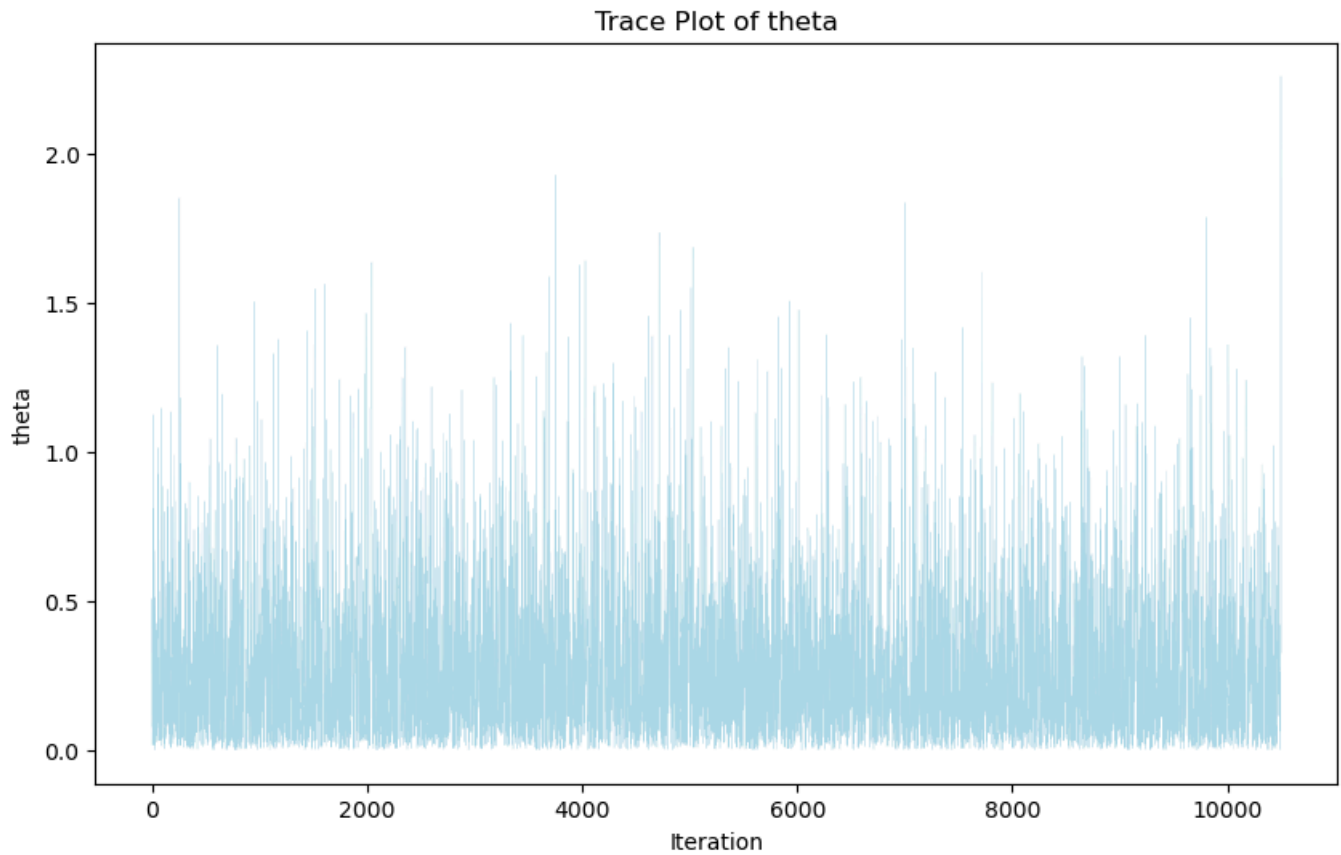
$$\pi(\theta|1/2, 1) = \frac{1^{\frac{1}{2}}}{\Gamma(1/2)} \theta^{-\frac{1}{2}} e^{-\theta}$$

$$\pi(\theta) \propto \theta^{-\frac{1}{2}} e^{-\theta}$$

Proposal distribution:

$$q(\theta', \theta) \propto \theta'^{\alpha-1} e^{-\beta\theta'}$$

After testing different a and b values, the initial values of $a = 1$ and $b = 4$ are chosen for now to showcase the Metropolis-Hastings workflow with a suitable acceptance criteria.



1b)

After burning the first 500 samples, the acceptance rate is around 0.86 for our Gamma posterior distribution. Although, as we have chosen $a = 1$ the distribution is equivalent to an exponential distribution.

In the case of a non exact distribution, take the mean of all our theta samples. For $a = 1, b = 4$, the Bayes estimator is ~ 0.33 .

HDI Credible Interval calculated using Arviz package: [0.00, 1.14]

1c)

Finding the exact posterior distribution using the likelihood and prior defined above:

$$\pi(\theta|X) \propto L(\theta|X) \cdot \pi(\theta)$$

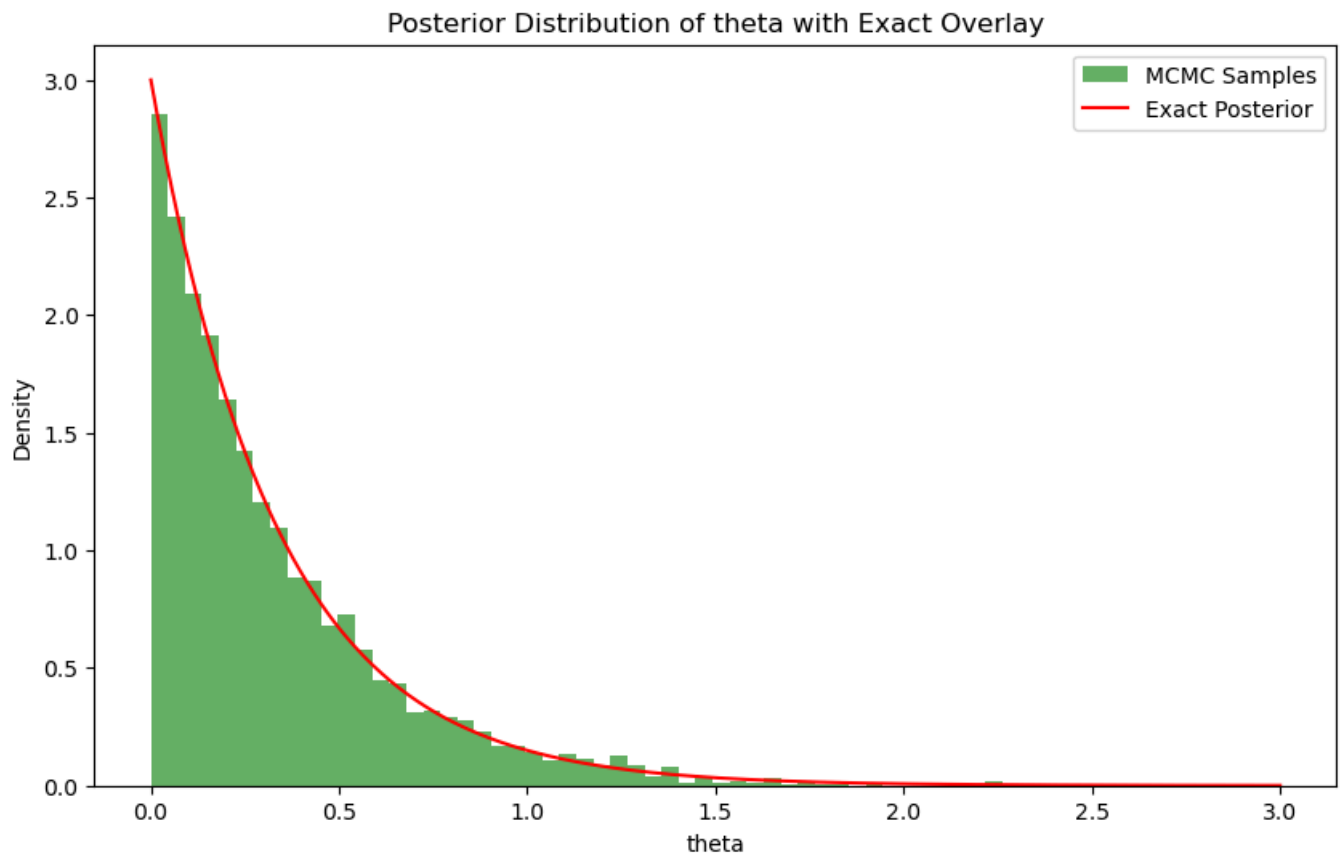
$$\pi(\theta|X) \propto (\theta^{\frac{1}{2}} e^{-2\theta}) (\theta^{-\frac{1}{2}} e^{-\theta})$$

Using the kernel method to compare indices:

$$\theta^{\alpha-1} e^{-\beta\theta} \propto (\theta^{\frac{1}{2}} e^{-2\theta}) (\theta^{-\frac{1}{2}} e^{-\theta})$$

$$\theta^{\alpha-1} e^{-\beta\theta} \propto e^{-3\theta}$$

$$\alpha = 1, \quad \beta = 3$$



The posterior distribution is a gamma function with $a=1$, which is equivalent to an exponential distribution.

1d)

Bayes estimator (posterior mean) for Metropolis-Hastings, given it's a exponential distribution and we know the exact distributions ($a=1$, $b=3$):

$$\mu_{posterior} = \frac{a}{b} = \frac{1}{3}$$

Acceptance Rate: 1

Bayes Estimator (posterior mean) of theta (after burn-in): 0.33

HDI Credible Interval (using ArviZ) for exact posterior: [0.0, 1.17]

As expected the accepted rate is 100% as this is an exact distribution.

For properties of the non-exact posterior distribution tested where $a = 1$, $b = 4$:

Acceptance Rate: 0.86

The Bayes estimator is ~ 0.33 .

HDI Credible Interval calculated using Arviz package: [0.00, 1.14]

Question 2

2a)

To find the conditional distribution first we need to compute the joint distribution, which would be the likelihood multiplied by the prior for both θ and y_0 .

Prior for θ :

$$\pi(\theta) \sim \text{Uniform}(0, 1)$$

Here the probability density function = 1 as, $\text{pdf} = \frac{1}{b-a} = \frac{1}{1-0}$.

We do not need to consider a prior for y_0 as it is a latent variable and not a parameter.

Joint distribution:

$$f(\theta, y_0, \bar{y}) = p(\bar{y}|\theta) \cdot \pi(\theta) \cdot \pi(y_0)$$
$$f(\theta, y_0, \bar{y}) \propto \frac{n!}{y_0!(y_1 - y_0)!y_2!y_3!y_4!} \left(\frac{1}{2}\right)^{y_0} \left(\frac{\theta}{4}\right)^{y_1 - y_0 + y_4} \left(\frac{1 - \theta}{4}\right)^{y_2 + y_3}$$

Conditional Distribution of y_0

Isolating just the terms with y_0 :

$$\pi(y_0|y_1, \theta) = \frac{y_1!}{y_0!(y_1 - y_0)!} \left(\frac{\theta}{4}\right)^{y_1 - y_0} \left(\frac{1}{2}\right)^{y_0}$$

Normalise with y_1 probability: $\left(\frac{1}{2} + \frac{\theta}{4}\right)^{y_1}$

$$\pi(y_0|\theta, y_1) = \binom{y_1}{y_0} \frac{\left(\frac{\theta}{4}\right)^{y_1 - y_0} \left(\frac{1}{2}\right)^{y_0}}{\left(\frac{1}{2} + \frac{\theta}{4}\right)^{y_1}} = \binom{y_1}{y_0} \frac{\left(\frac{\theta}{4}\right)^{y_1 - y_0} \left(\frac{1}{2}\right)^{y_0}}{\left(\frac{1}{2} + \frac{\theta}{4}\right)^{y_0} \cdot \left(\frac{1}{2} + \frac{\theta}{4}\right)^{y_1 - y_0}}$$
$$\pi(y_0|\theta) = \binom{y_1}{y_0} \left(\frac{\frac{\theta}{4}}{\frac{1}{2} + \frac{\theta}{4}}\right)^{y_1 - y_0} \left(\frac{\frac{1}{2}}{\frac{1}{2} + \frac{\theta}{4}}\right)^{y_0}$$

Setting $\phi = \frac{\frac{1}{2}}{\frac{1}{2} + \frac{\theta}{4}}$ and using the hint provided we can state:

$$\pi(y_0|\theta, y_1) = \binom{y_1}{y_0} \phi^{y_0} (1 - \phi)^{y_1 - y_0}$$
$$\pi(y_0|\theta, y_1) \sim \text{Bin}(y_1, \phi)$$

Conditional Distribution of θ

Isolating just the terms with θ :

$$\pi(\theta|y_0, \bar{y}) \propto \left(\frac{\theta}{4}\right)^{y_1 - y_0 + y_4} \left(\frac{1 - \theta}{4}\right)^{y_2 + y_3}$$

We can bring the $\frac{1}{4}$ term out and it becomes a constant which can be ignored.

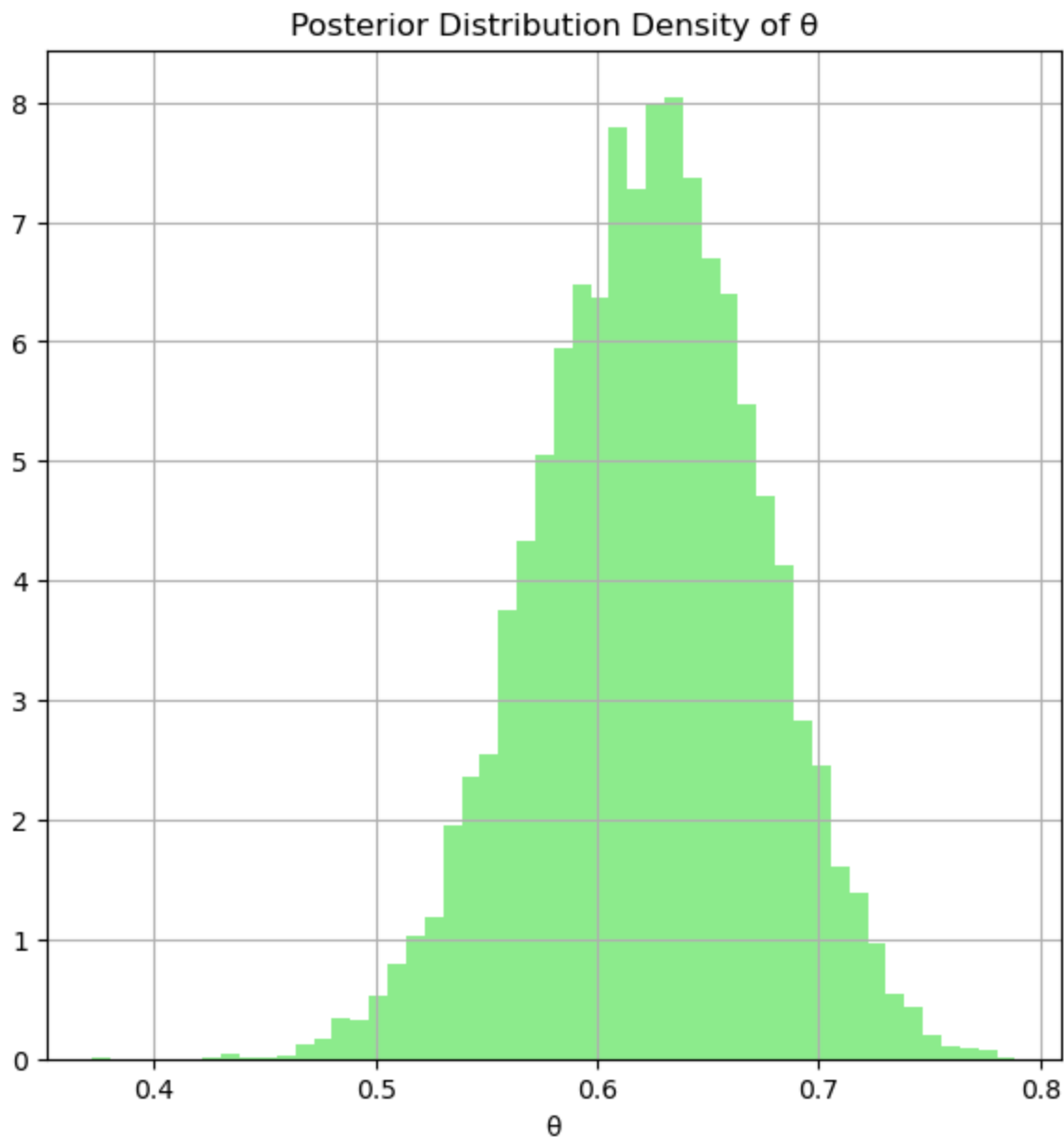
$$\pi(\theta|y_0, \bar{y}) \propto \theta^{y_1 - y_0 + y_4} (1 - \theta)^{y_2 + y_3}$$

This resembles a Beta distribution:

$$\pi(\theta|y_0, \bar{y}) \sim \text{Beta}((y_1 - y_0 + y_4 + 1), (y_2 + y_3 + 1))$$

where $\alpha = (y_1 - y_0 + y_4 + 1)$, $\beta = (y_2 + y_3 + 1)$.

2b)



2c)

Point Estimate (posterior mean) of theta: 0.622

95% equi-tailed credible interval: [0.52, 0.72]

Probability within set: 0.95

Question 3

3a)

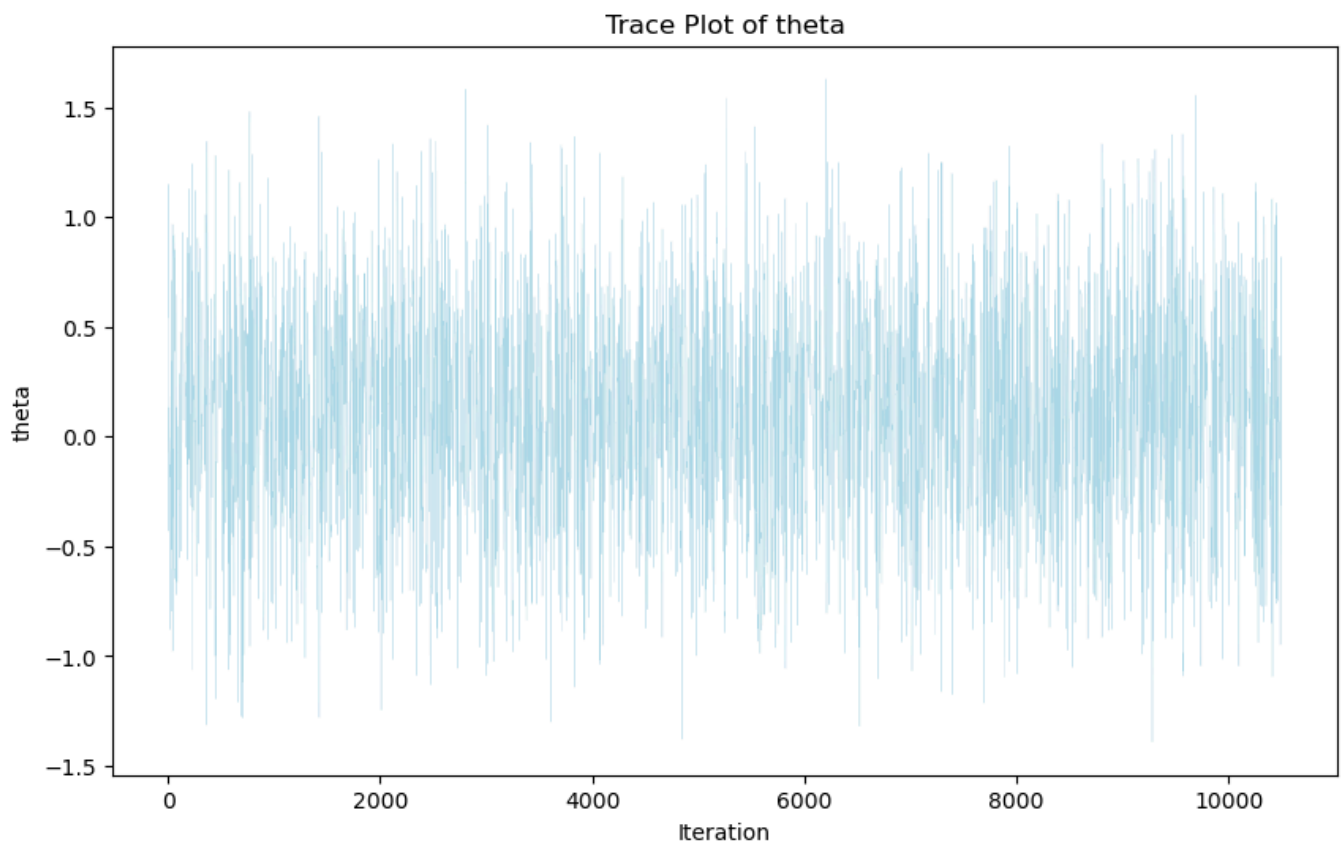
Given the likelihood equation, we know its distribution is the following:

$$f(y|\theta) \sim N(\theta, \frac{1}{\tau})$$

The Proposal distribution is assumed to be a uniform distribution between -2 and 2. Considering we want a stationary distribution with specific bounds, a uniform distribution is the simplest to implement, although a truncated normal distribution could also be an interesting proposal distribution given it meets our criteria.

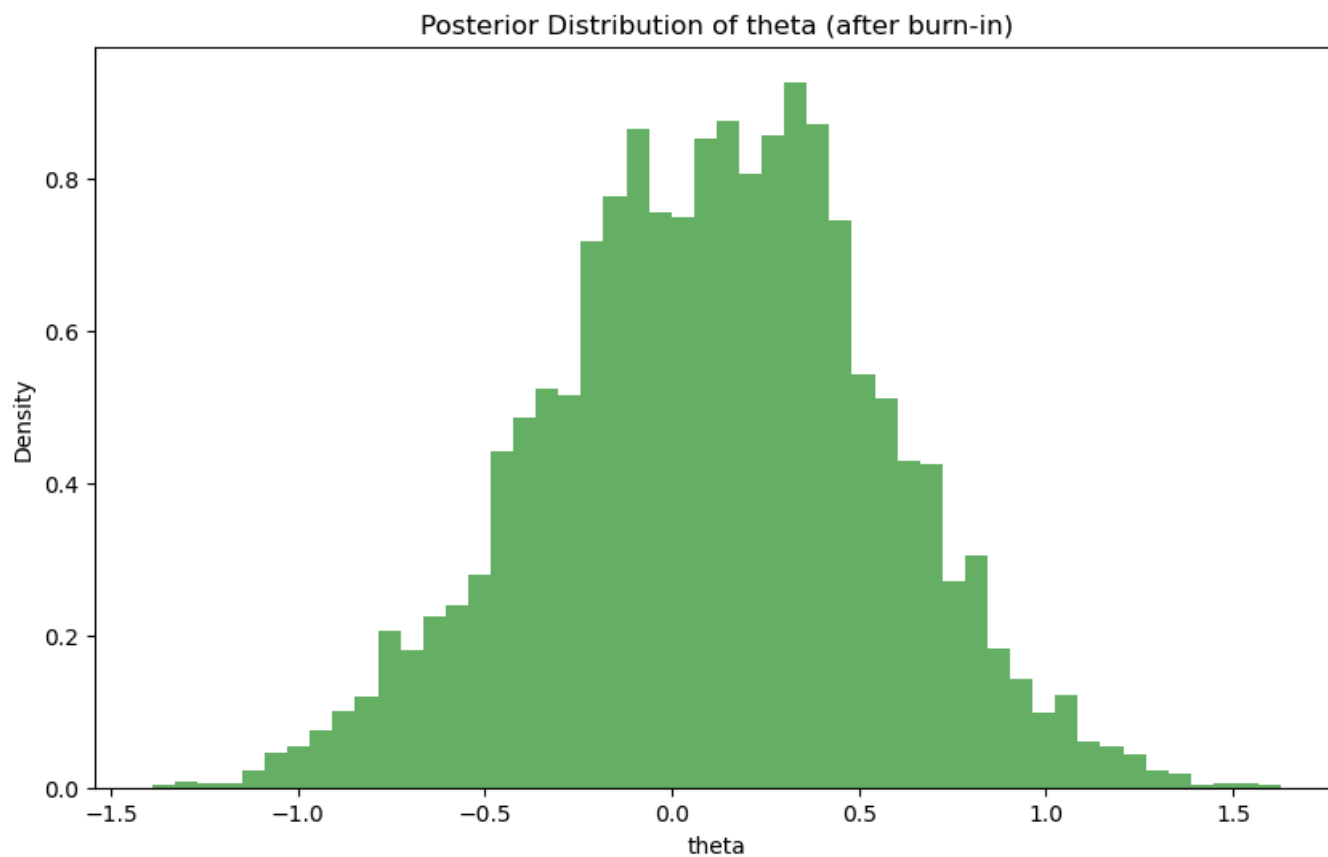
$$q(\theta', \theta) \sim Uniform(-2, 2)$$

3b)



3c)

After burning first 500 samples, the histogram resembles a normal distribution.



3d)

Bayes estimator (posterior mean) of theta (after burn-in): 0.108

95% Equi-tailed Credible Interval: [-0.80, 0.97]

Probability within set: 0.95